

Week 4: Machine Learning Model Development and Evaluation

Breast Cancer Classification Using Machine Learning

Virtual Data Science with Python Apprentice Internship

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1. Introduction

This project focuses on developing and evaluating machine learning models using Python. The Breast Cancer dataset available in Scikit-learn was used to build classification models that predict whether a tumor belongs to the malignant or benign class. The workflow included data exploration, preprocessing, model training, prediction, performance evaluation, and comparison of different machine learning algorithms.

2. Dataset Description

The Breast Cancer dataset contains numerical features calculated from breast mass samples. The target variable has two classes: 0 represents malignant and 1 represents benign.

3. Data Preparation

The dataset was loaded directly from Scikit-learn and converted into a Pandas DataFrame. It was checked for missing values, duplicate records, and class distribution. The data was split into training and testing sets using an 80:20 ratio with stratification. StandardScaler was applied for Logistic Regression because the features have different numerical ranges.

4. Model Selection

Two classification algorithms were selected. Logistic Regression was used as a simple and interpretable baseline model for binary classification. A Decision Tree Classifier was selected because it can capture non-linear relationships and does not require feature scaling.

5. Model Training

Logistic Regression was trained using scaled training features, while the Decision Tree was trained using the original feature values. Predictions were then generated for the unseen test dataset.

6. Model Evaluation

The models were evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC Curve, and Area Under the Curve (AUC). These metrics provide a broader understanding of model performance.

7. Visualization 1: Confusion Matrix

Confusion matrices were created for both Logistic Regression and Decision Tree models. Insert screenshots of the confusion matrices generated in Google Colab in this section.

8. Visualization 2: ROC Curve

A ROC curve was created to compare the classification performance of both models across different thresholds. Insert the ROC curve screenshot generated in Google Colab in this section.

9. Visualization 3: Model Accuracy Comparison

A bar chart was created to compare the accuracy of Logistic Regression and Decision Tree models. Insert the accuracy comparison chart in this section.

10. Results

Copy your actual results from the Colab notebook here.

Logistic Regression Accuracy: _____
Logistic Regression Precision: _____
Logistic Regression Recall: _____
Logistic Regression F1-score: _____
Logistic Regression AUC: _____

Decision Tree Accuracy: _____
Decision Tree Precision: _____
Decision Tree Recall: _____
Decision Tree F1-score: _____
Decision Tree AUC: _____

11. Discussion

The final model comparison should be based on the evaluation metrics obtained during testing. Accuracy alone should not be the only criterion. Precision, recall, F1-score, ROC-AUC, and the confusion matrix provide additional information about model strengths and errors.

12. Potential Sources of Error and Improvements

Possible limitations include the dataset characteristics, the selected train-test split, model hyperparameters, and the possibility of overfitting. Future improvements could include hyperparameter tuning, cross-validation, feature selection, Decision Tree pruning, and testing additional algorithms such as Random Forest or Support Vector Machines.

13. Conclusion

This project demonstrated a complete machine learning workflow using Python and Scikit-learn. Logistic Regression and Decision Tree models were developed to classify tumors as malignant or benign. Their performance was evaluated using multiple metrics and visualizations. The project provided practical experience in data preprocessing, model training, prediction, performance evaluation, and identifying possible improvements.